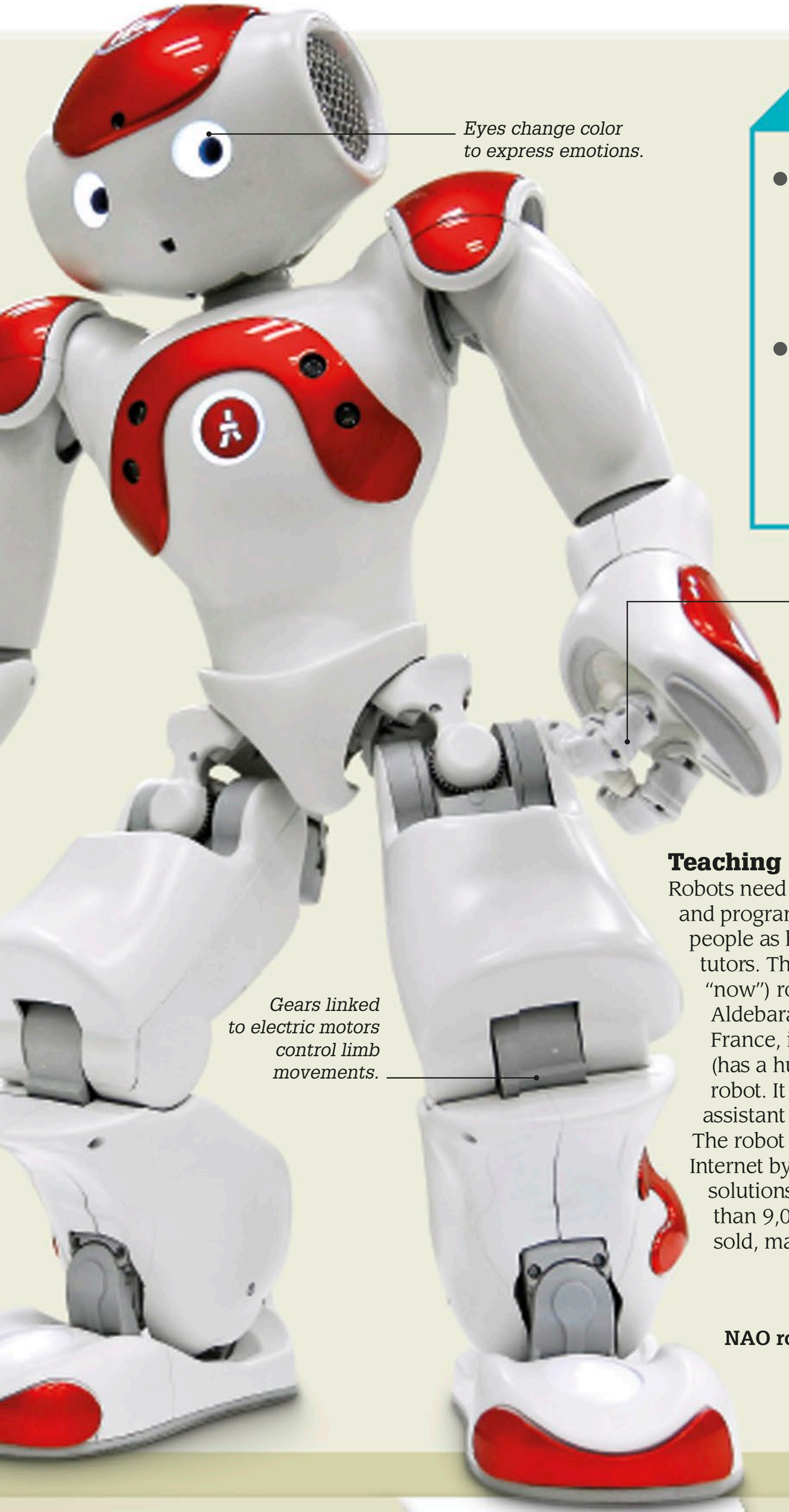


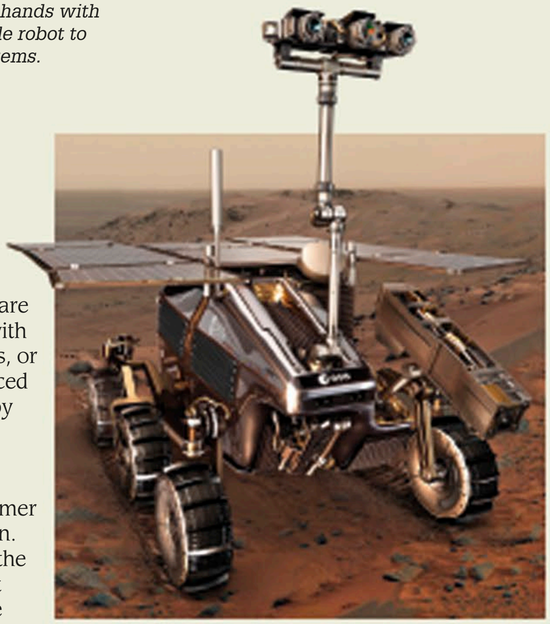
Key robotic components

- **Controller**
  The computer software and hardware that act as the brain of the robot, making decisions and instructing a robot's parts.
- **End effectors**
  The parts of a robot that interact with its surroundings. These may include a gripper on an arm, which holds objects.
- **Sensors**
  Devices such as cameras, distance detectors, and GPS (Global Positioning System), which gather data for the controller.
- **Drive system**
  The system used to power a robot's moving parts. It is usually electrical, pneumatic (operated by compressed gases), or hydraulic (operated by compressed liquid).

Teaching robots

Robots need advanced hardware and programming to work with people as helpers, assistants, or tutors. The NAO (pronounced "now") robot, developed by Aldebaran Robotics in France, is a humanoid (has a humanlike body) robot. It works as a customer assistant at a bank in Japan. The robot is able to link to the Internet by itself to seek out solutions to queries. More than 9,000 NAOs have been sold, mainly in education.

NAO robot, 2015

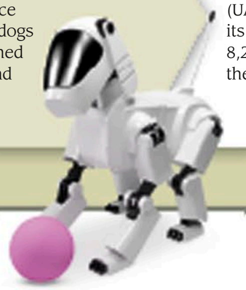


Robotic explorers

Space robots, such as the ExoMars rover (to be launched to Mars as early as 2020), explore places too hostile for humans—such as the Martian surface—and send findings back to Earth using radio signals. Robots explore Earth as well, from inside the narrow shafts of Ancient Egyptian pyramids to the deepest ocean beds.

1999

Sony launched AIBO (Artificial Intelligence Bot). These robotic dogs could be programmed to perform tasks and became popular in education.



2001

An unmanned aerial vehicle (UAV) called Global Hawk plotted its own course as it flew about 8,214 miles (13,219 km) from the US to Australia.

2011

The Robonaut 2 humanoid robot was sent to the International Space Station. There, the two-armed robot was tested performing repetitive tasks, such as cleaning air filters.

2014

Roomba, a line of robotic vacuum cleaners, became the most common robots in the world. Since their launch in 2002, more than 10.5 million Roombas had been sold.